



Establishment of Full-Length cDNA Clones and an Efficient Oral Infection Model for Feline Coronavirus in Cats

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ABSTRACT Feline infectious peritonitis virus (FIPV) is the etiologic agent of feline infectious peritonitis (FIP) and causes fatal disease in cats of almost all ages. Currently, there are no clinically approved drugs or effective vaccines for FIP. Furthermore, the pathogenesis of FIP is still not fully understood. There is an urgent need for an effective infection model of feline infectious peritonitis induced by FIPV. Here, we constructed a field type I FIPV full-length cDNA clone, pBAC-QS, corresponding to the isolated FIPV QS. By replacing the FIPV QS spike gene with the commercially available type II FIPV 79-1146 (79-1146_CA) spike gene, we established and rescued a recombinant virus, designated rQS-79. Moreover, we constructed 79-1146_CA infectious full-length cDNA pBAC-79-1146_CA, corresponding to recombinant feline coronavirus (FCoV) 79-1146_CA (r79-1146_CA). In animal experiments with 1- to 2-year-old adult cats orally infected with the recombinant virus, rQS-79 induced typical FIP signs and 100% mortality. In contrast to cats infected with rQS-79, cats infected with 79-1146_CA did not show obvious signs. Furthermore, by rechallenging rQS-79 in surviving cats previously infected with 79-1146_CA, we found that there was no protection against rQS-79 with different titers of neutralizing antibodies. However, high titers of neutralizing antibodies may help prolong the cat survival time. Overall, we report the first reverse genetics of virulent recombinant FCoV (causing 100% mortality in adult cats) and attenuated FCoV (causing no mortality in adult cats), which will be powerful tools to study pathogenesis, antiviral drugs, and vaccines for FCoV.

IMPORTANCE Tissue- or cell culture-adapted feline infectious peritonitis virus (FIPV) usually loses pathogenicity. To develop a highly virulent FIPV, we constructed a field isolate type I FIPV full-length clone with the spike gene replaced by the 79-1146 spike gene, corresponding to a virus named rQS-79, which induces high mortality in adult cats. rQS-79 represents the first described reverse genetics system for highly pathogenic FCoV. By further constructing the cell culture-adapted FCoV 79-1146_CA, we obtained infectious clones of virulent and attenuated FCoV. By *in vitro* and *in vivo* experiments, we established a model that can serve to study the pathogenic mechanisms of FIPV. Importantly, the wild-type FIPV replicase skeleton of serotype I will greatly facilitate the screening of antiviral drugs, both *in vivo* and *in vitro*.

KEYWORDS feline infectious peritonitis virus, pathogenesis, reverse genetics

Feline coronavirus (FCoV) is classified in the *Alphacoronavirus* genus of the *Coronaviridae* family and is generally divided into feline enteric coronavirus (FECV) and feline infectious peritonitis virus (FIPV) according to biotype or pathogenicity (1–3). FECV infection is generally endemic to domestic cat populations worldwide and causes either mild enteritis or subclinical infections. In contrast to FECV infection, FIPV infection results fatal disease in

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cats once clinical signs develop. Fortunately, FIPV has rare to no capacity for horizontal transmission from cat to cat (1).

FCoVs are widespread and divided into serotypes I and II according to the spike gene. Most natural FCoV infections are caused by serotype I, which accounts for 70% to 98% of FCoV infections worldwide (4–10). There is consistent evidence that serotype II FCoVs evolved from partial RNA sequence recombination containing the spike gene of both serotype I FCoVs and canine coronaviruses (CCoVs) (10–12). Two biotypes, FECV and FIPV, designated according to their different pathogenicities in cats, exist as serotype I or II (10). It is still difficult to distinguish between the nonvirulent FECV and the highly virulent FIPV at the genomic sequence level (13). Although some research has stated that FIPV can be distinguished from FECV by spike protein amino acid mutations, these mutations have since been found to be more related to tissue tropism (13, 14).

Reverse genetics has been a powerful tool for the study of CoV vaccines and pathogenesis (1, 15). In recent years, several FCoVs, including moderately virulent FCoV reverse genetics systems, have been established, but highly lethal and cell culture-grown FCoV has not yet been developed, especially for type I FCoV (16–19).

In this study, we constructed a full-length genome of serotype I field isolates of FIPV QS, which was isolated first in our lab. By replacing the spike gene of the FIPV QS field isolate with that of 79-1146_CA, we obtained a complete field FIPV skeleton recombinant virus with the serotype II spike gene, named rQS-79, which grows well in CRFK or FCWF-4 cells. Furthermore, we developed a nonvirulent FCoV infectious clone from the cell culture-adapted feline CoV 79-1146, corresponding to a virus named r79-1146_CA. By experimentally infecting adult cats orally with the r79-1146_CA or rQS-79 recombinant virus, we found that rQS-79 induced 100% mortality, while no mortality was found with r79-1146_CA infection. The establishment of a model of highly virulent and attenuated FCoV infections in cats will greatly promote research on the pathogenic mechanism of FCoV. By rechallenge protection experiments in animals orally infected with highly virulent and attenuated FCoVs, we further found that different titers of neutralizing antibodies (NAs) did not completely protect the cats against rQS-79 challenge, while high titers of NAs prolonged the cat survival time. Furthermore, the FCoV candidate drug, GS-441524, can be applied to artificially induced FIP cats with a good therapeutic effect.

RESULTS

Full-length genome sequence of the serotype I FIPV field isolate and cell-adapted serotype II FIPV 79-1146. To obtain a highly virulent FCoV field isolate, ascites samples from a cat with clinically confirmed FIP were collected. By overlapping reverse transcription-PCR fragments and 5'–3' rapid amplification of cDNA ends (RACE) sequencing, we determined the full-length genome sequence of FCoV from the FIP cat. The field FCoV was designated QS (GenBank accession no. [MW030108](#)), which was verified as a serotype I FCoV by comparison with the type I FCoV strain Black S gene (GenBank accession no. [EU186072](#); 88.43% amino acid identity) and the type II FCoV strain 79-1146 S gene (GenBank accession no. [DQ010921](#); 44.92% amino acid identity). The QS genome had a genome organization typical of FCoVs (Fig. 1A). Additionally, we also determined the genome of a commercially available cell-cultured serotype II FIPV 79-1146 by overlapping reverse transcription-PCR fragment sequencing (Fig. 1B). We designated the commercially available cell-cultured FCoV 79-1146_CA (GenBank accession no. [MW030109](#)), which was passaged over 30 times after being obtained from the American Type Culture Collection (ATCC).

Construction and recovery of the recombinant FCoV QS and 79-1146_CA. Since viruses are difficult to isolate directly from samples containing FCoVs, we used reverse genetics to directly rescue FCoV QS. To construct a full-length genome clone of FCoV QS in a bacterial artificial chromosome (BAC), the QS genome was divided into eight fragments, designated A to H (Fig. 1A). All fragments were generated by reverse transcriptase PCR (RT-PCR) using total RNA extracted from ascites samples from a cat with

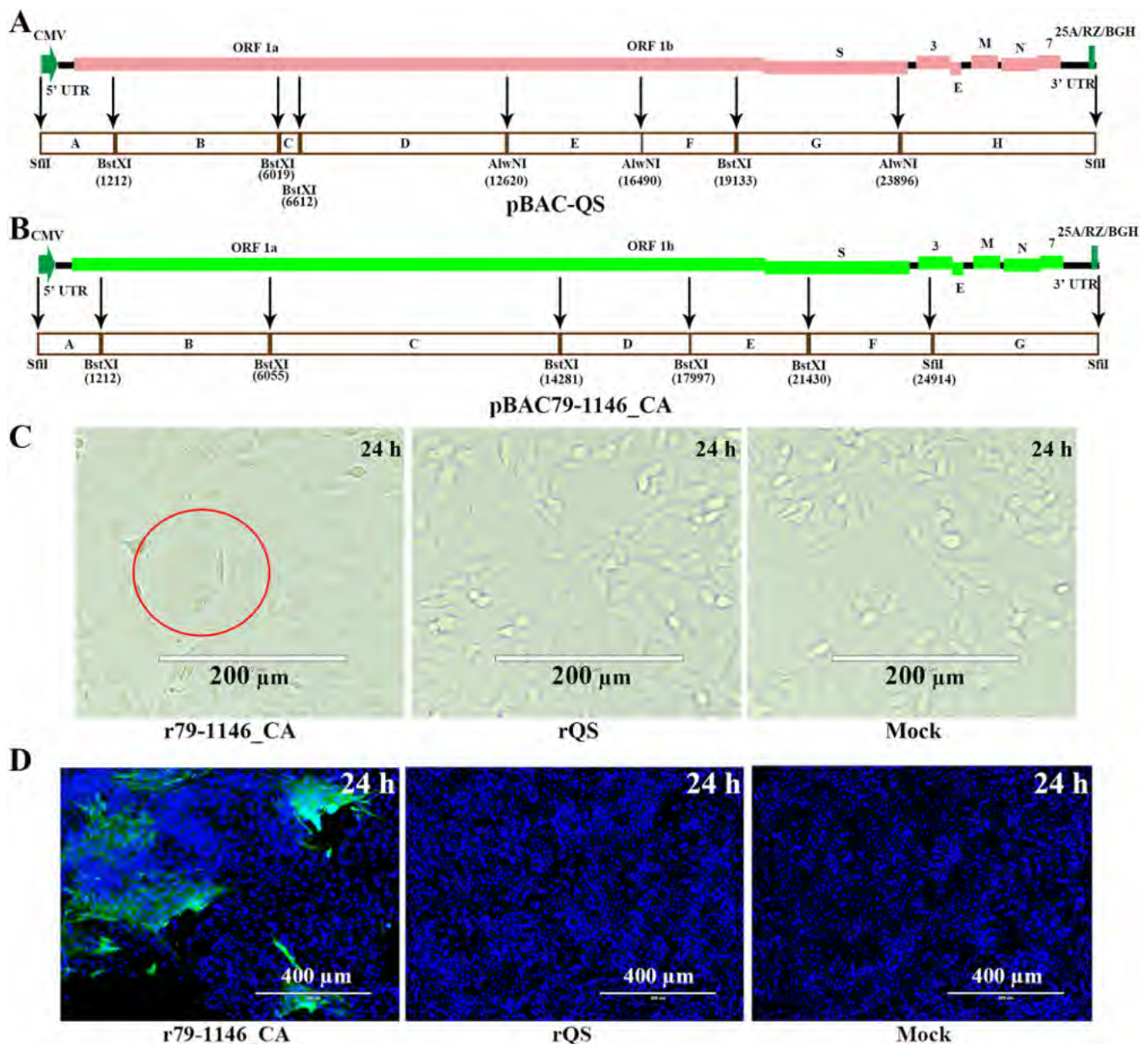


FIG 1 Construction and recovery of recombinant FCoV QS and 79-1146_CA full-length cDNA clones. (A) Schematic diagram showing the strategy for constructing the FCoV QS full-length cDNA clone in the BAC vector. The full-length QS genome was divided into eight fragments (A to H) by AlwNI, SfiI, and BstXI sites. The eight fragments, along with the CMV promoter, HDV ribozyme, and BGH transcription terminal signal sequences, then were inserted into pBAC-QS in one step. (B) Schematic diagram showing the strategy for constructing the FCoV 79-1146_CA full-length cDNA clone in the BAC vector. The full-length 79-1146_CA genome was divided into seven fragments (A to G). Two natural BstXI sites were removed by introducing the mutations C644T and G6641G, which were maintained as rescue markers. (C) Recovery of recombinant FCoV QS and 79-1146_CA. Recombinant QS or 79-1146_CA full-length cDNA clones in the BAC vector were transfected into CRFK cells for rescue. An obvious CPE was observed at 24 h posttransfection. The red circle indicates the CPE of syncytia. (D) Indirect IFA for detecting FCoV in transfected CRFK cells. Green fluorescence indicates the N protein expression of recovered recombinant FCoV.

clinically confirmed FIP. Fragments A, B, C, E, F, G, and H were cloned into pMD-18T, while fragment D was cloned into the BAC to stabilize the subclone. Fragment A was constructed by fusion with the cytomegalovirus (CMV) promoter at the 5' terminus, while fragment G was constructed by fusion with a 25-bp poly(A) tail (25A) followed by hepatitis delta virus (HDV) ribozyme (Rz) and bovine growth hormone (BGH) transcription terminal signal sequences at the 3' terminus (Fig. 1A). Through one-step assembly of fragments A to H with the BAC vector, we successfully obtained a full-length cDNA clone of QS, designated pBAC-QS. To construct an infectious clone of FCoV 79-1146_CA,

the 79-1146_CA genome was divided into seven fragments, designated A to G (Fig. 1A). Fragments A, B, D, E, F, and G were cloned into pMD-18T, while fragment C was cloned into the BAC to stabilize the subclone (Fig. 1B). The other elements for 79-1146_CA construction were similar to those for QS. Through one-step assembly of fragments A to G with the BAC vector, we successfully obtained a full-length cDNA clone of 79-1146_CA in the BAC, designated pBAC79-1146_CA (Fig. 1B).

To rescue recombinant FCoV r79-1146_CA and rQS (corresponding to pBAC79-1146_CA and pBAC-QS, respectively), pBAC79-1146_CA or pBAC-QS was transfected into CRFK cells using Lipofectamine 2000. After 24 h of pBAC-79-1146_CA transfection, an obvious cytopathic effect (CPE) of syncytial fusion was observed (Fig. 1C). The mock and pBAC-QS transfection groups did not show any CPE, which was verified by indirect immunofluorescence assays (IFAs) (Fig. 1D). These data indicated that serotype I FIPV field QS cannot be propagated in standard cell culture systems.

Recovery of recombinant QS with the serotype II FIPV 79-1146_CA S gene. To identify the complete functional activity of the field serotype I FIPV QS genome, except the S gene lacking its transmembrane and cytoplasmic domains (Δ TCD), we rescued the recombinant QS with serotype II FIPV 79-1146_CA Δ TCD replacing FIPV QS Δ TCD, designated pBAC-QS-79 (Fig. 2A). Using the CRISPR-Cas9 system *in vitro*, we efficiently digested pBAC-QS to yield a linearized BAC and an approximately 4.3-kb fragment that included the sequence of FIPV QS Δ TCD (Fig. 2B). To construct the recombinant QS with the serotype II FIPV 79-1146_CA Δ TCD gene, the mutation PCR products of FIPV 79-1146_CA Δ TCD replacing FIPV QS Δ TCD were recombined into digested purified pBAC-QS. To identify recombinant pBAC-QS-79, the positive BAC plasmid was digested with HindIII and sequenced (Fig. 2C and D). After transfection of pBAC-QS-79 into CRFK cells, the obvious CPE and IFA results also demonstrated the recovery of the rQS-79 recombinant virus (corresponding to pBAC-QS-79) (Fig. 2D and E). To characterize rQS-79 and r79-1146_CA, we measured their growth kinetics. During the 8 to 32 h after inoculation, the titer of rQS-79 was approximately 10 times less than that of r79-1146_CA (Fig. 2F). The immunofluorescent plaque sizes for rQS-79 and r79-1146_CA were compared at 48 h after inoculation. r79-1146_CA plaques were more than 2 times larger than rQS-79 plaques in diameter (Fig. 2G). These results indicated that the field serotype I FIPV QS genome, except Δ TCD, functions completely and that the sequence of Δ TCD obstructs field serotype I FIPV recovery or isolation in CRFK cells.

Challenge of cats with rQS-79 or r79-1146_CA. To analyze the pathogenicity of rQS-79 and r79-1146_CA in cats, 25 1- to 2-year-old cats were randomly divided into three groups, with 10 cats in the rQS-79 group, 10 cats in the r79-1146_CA group, and 5 cats in the mock group. Cats in the rQS-79 group and r79-1146_CA group were orally inoculated with 1 ml of rQS-79 and r79-1146_CA at $1 \times 10^{5.5}$ 50% tissue culture infectious dose (TCID₅₀)/ml, respectively. Cats in the mock group were orally inoculated with 1 ml of cell culture Dulbecco's modified Eagle's medium (DMEM). In the rQS-79 group, all cats showed obvious clinical signs, including fever, weight loss, anorexia, depression, and rough back hair. Specifically, all cats in the rQS-79 group developed a persistent fever several days after virus challenge, while cat body temperature in the r79-1146_CA group returned to normal within 1 week compared to that of the mock group (Fig. 3A). The clinical scores of cats in the rQS-79 group were significantly worse than those of cats in the r79-1146_CA or mock group (Fig. 3B). The average relative weight of cats in the rQS-79 group (87% starting weight) was significantly lower than that of cats in the r79-1146_CA (97% starting weight) or mock (102% starting weight) group (Fig. 3C). Like the cats in the mock group, none of the cats in the r79-1146_CA group died within the 2-month observation period, except for one cat that was euthanized to allow a comparative assessment for pathological lesions consistent with or indicative of FIP, but all the rQS-79 group cats died (eight dying cats with severe dyspnea and/or inability to stand were euthanized, and two cats died naturally) within 18 days (Fig. 3D). Additionally, to assess virus shedding, oral and rectal swab samples were collected from each cat at 3 and 6 days postinfection. Oral and rectal swab samples were tested for the presence of FCoV by RT-PCR for the S gene and inoculated into CRFK

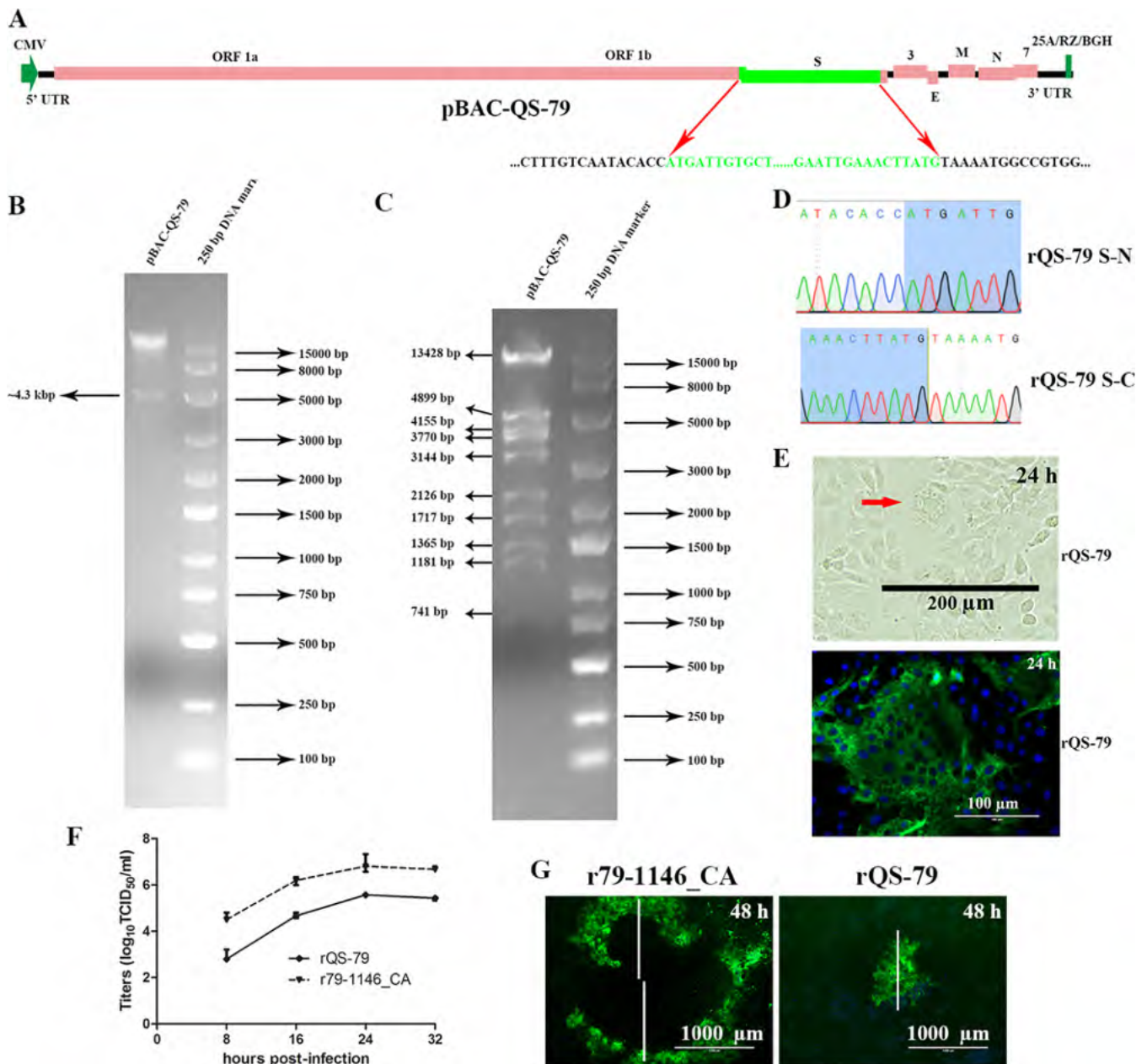


FIG 2 Recovery of rQS-79 and characterization of rQS-79 and r79-1146_CA. (A) Schematic diagram showing the strategy for constructing rQS-79. The 79-1146_CA Δ TCD sequence is shown in green. (B) Specific cleavage of pBAC-QS to delete the sequence containing Δ TCD. The pBAC-QS clones digested by Cas9 guided by sgRNA were detected by electrophoresis. (C) The left lane is the product of HindIII digestion of pBAC-QS-79, and the right lane is the DL15000 DNA marker. (D) Site analysis of 79-1146_CA Δ TCD replacing QS Δ TCD by sequencing. (E) Recovery of the recombinant FCoV rQS-79. rQS-79 full-length cDNA clones in the BAC vector were transfected into CRFK cells for rescue. Obvious cytopathic effects (CPEs) and antigen signals identified by IFA were observed at 24 h posttransfection. (F) Growth curves of recombinant FCoV rQS-79 and r79-1146_CA with an original MOI of 0.01. (G) Viral fluorescent plaques identified by IFA for rQS-79 and r79-1146_CA. The white vertical bars have the same length.

cells to assess the infectious virus titers. Interestingly, only three oral samples were positive for the virus S gene (Fig. 3E), and two samples contained low-titer infectious viruses when inoculated into CRFK cells (Fig. 3F and G).

Post mortem examination of cats infected with recombinant FCoV. After 18 days of recombinant FCoV (rFCoV) inoculation, the last two moribund cats were euthanized in the rQS-79 group, while two cats, one selected at random in the r79-1146_CA group (cat 10) and one from the mock group (cat 21), were also euthanized for a comparative assessment. At necropsy, all 10 cats in the rQS-79 group were positive for rQS-79 according to the RT-PCR assay (data not shown). Two cats (2/10) in the

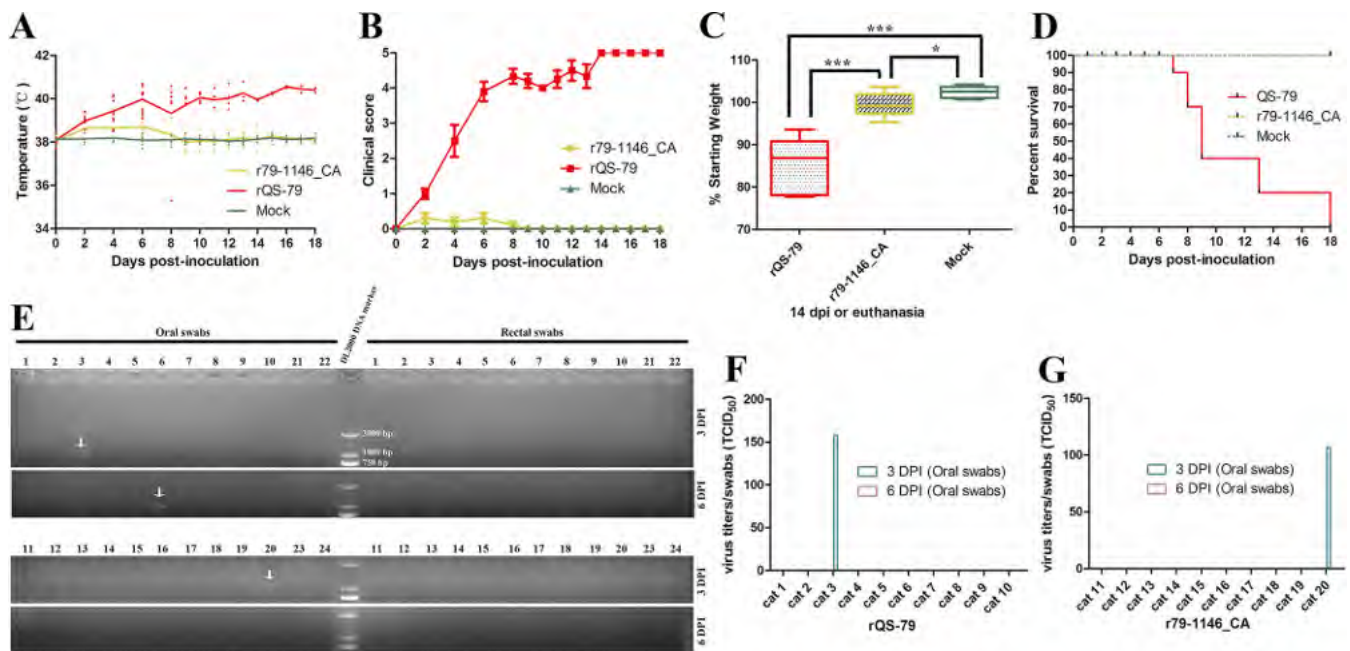


FIG 3 Pathogenicity and transmission evaluation of recombinant viruses. (A) The cat body temperatures responded to viral infection over time. (B) The clinical status scores of the cats (in any of the following cases, 1 point will be added) were the following: 1, cats that had a fever ($>39^{\circ}\text{C}$); 2, cats that were anorexic; 3, cats that were lethargic; 4, cats that had weight loss ($>5\%$); and 5, cats that were unable to stand or severe dyspnea and were extremely sick. (C) Percent survival of the different cats in each group. (D) Average relative weight at the time of euthanasia or 14 days after inoculation. On day 0 before challenge, the body weight was 100%. (E) Recombinant FCoV nucleic acids were detected in all cats by virus-specific RT-PCR on oral and rectal swabs at 3 or 6 days postinoculation. The white arrow indicates the recombinant FCoV nucleic acid. Numbers 1 to 24 above agarose gel wells represent the corresponding cat (rQS-79 group, cats 1 to 10; r79-1146_CA group, cats 11 to 20; and mock group, cats 21 to 24). No recombinant FCoV nucleic acid was detected in cat 25 (data not shown). (F and G) Titer determination of live recombinant virus was performed in CRFK cells at 3 or 6 days postinoculation. Asterisks indicate significant differences between rQS-79 and mock (***, $P < 0.0005$), between rQS-79 and r79-1146_CA (***, $P < 0.0005$), and between r79-1146_CA and mock (*, $P < 0.05$). The error bars indicate standard deviations.

rQS-79 group showed thoracic/abdominal effusions, and 8 cats (8/10) showed gross lesions consistent with FIP, such as variably sized parenchymal and serosal pyogranulomas in affected organs (Table 1). In these FIP cats, a fibrinous to granulomatous serositis affected the spleen, liver, and kidney (Fig. 4A and B). In the hematoxylin and eosin (HE) staining analysis of different tissue sections, granuloma-related lesions are surrounded by dense inflammatory infiltrates. For example, distinct focal accumulations of macrophages were observed just under the capsule of the liver, spleen, and kidney. These foci were surrounded by dense accumulations of lymphocytes and plasma cells extending downward into the parenchyma (Fig. 4C). The immunofluorescence results further confirmed that viral antigen was detected within macrophages in the lesions of cats from the rQS-79 group (Fig. 4D).

Protective efficacy against rechallenge with rQS-79 in r79-1146_CA group cats that recovered from the first challenge. Following the initial viral infection, NA titers were detected in experimental cats. To evaluate the dynamic production of serum NA titers, serum samples collected from cats in the rQS-79 and r79-1146 groups were tested by the NA assay. Seroconversion of the animals started at approximately 7 days postinfection and subsequently increased rapidly in most cats (Fig. 5A and B). We found that the NA levels at approximately 4 weeks in the cats began to decline but remained at a relatively high level (Fig. 5B). To further verify whether NAs are protective against highly virulent FCoV, the surviving 9 cats in the r79-1146_CA group were divided into three groups: three cats in the rQS-79 (cats 1, 4, and 6) group, three cats in the r79-1146_CA (cats 3, 5, and 8) group, and three cats in the mock (cats 2, 7, and 9) group, with similar average antibody levels. Cats in the rQS-79 group were orally inoculated with 2 ml of rQS-79 virus at $1 \times 10^{5.5}$ TCID₅₀/ml, and cats in the r79-1146_CA group were orally inoculated with 2 ml of r79-1146_CA virus at $1 \times 10^{5.5}$ TCID₅₀/ml. Cats in the mock group were orally inoculated with 2 ml of cell culture DMEM. We

TABLE 1 Main details of the clinical signs and organ distribution of lesions in cats orally infected with rQS-79

Cat	Clinical signs	Organ distribution of lesions	Form of disease
1	Fever, wt loss, inappetence	Kidney, lung, liver	Noneffusive
4	Fever, wt loss, inappetence	Lung, small intestine, kidney, liver, spleen	Noneffusive
6	Fever, wt loss, ataxia, hindlimb paralysis, inappetence	Liver	Noneffusive
11	Fever, wt loss, inappetence	Spleen, liver, kidney, lung, ascite, Serosa of small intestine	Effusive
12	Fever, wt loss, inappetence	Ascite, kidney, lung, Serosa of small intestine	Effusive
13	Fever, wt loss, inappetence	Spleen, liver,	Noneffusive
14	Fever, wt loss, inappetence	Liver, spleen, kidney, lung	Noneffusive
15	Fever, wt loss, inappetence	Lung, kidney	Noneffusive
16	Fever, wt loss, inappetence	Lung, spleen, Serosa of small intestine	Noneffusive
17	Fever, wt loss, inappetence	Lung, kidney	Noneffusive
18	Fever, wt loss, inappetence	Spleen, liver,	Noneffusive
19	Fever, wt loss, inappetence	Spleen, liver, kidney	Noneffusive
20	Fever, wt loss, inappetence	Lung, liver	Noneffusive
T5	Fever, wt loss, inappetence	Spleen, liver, kidney	Noneffusive
T6	Fever, wt loss, inappetence	Lung, liver, lung	Noneffusive

found that all cats in the rQS-79 group showed clinical signs identical to cats with FIP in the first experiment, while no cats in the r79-1146_CA or mock group showed clinical signs (Fig. 5C and D). All three cats in the rQS-79 group died with noneffusive FIP (cats with severe dyspnea and/or inability to stand were euthanized) within 46 days of rechallenge (Fig. 4E). Interestingly, cat 6 seems recovered after 10 days of rechallenge, but this cat eventually had obvious neurological signs and died after 46 days of rechallenge (Fig. 5C to E). The cats in the rQS-79 group showed obvious weight loss (Fig. 5F). At necropsy, we found that cats 1 and 4 that underwent reinoculation with rQS-79 had pathological lesions of FIP, especially in the kidney, showing numerous granuloma-related lesions (Fig. 5G). Overall, no cats in the r79-1146_CA or mock group with a second challenge developed clinical signs within more than 4 months of observation.

To analyze the histological changes in cats rechallenged with rQS-79, we used the kidneys as examples and observed that lesions of noneffusive FIP tended to extend downward from the serosal into underlying parenchyma (Fig. 5G). Typical FIP perivascularitis in the kidney was present, while distinct focal accumulations of macrophages, lymphocytes, and plasma cell infiltrates were observed under the capsule of the kidney (Fig. 6A). The immunofluorescence results further confirmed that viral antigen was detected within macrophages in the lesions (Fig. 6B). In the HE staining analysis of the specific pathological changes in the brains of cats rechallenged with rQS-79, we found the presence of a typical FIP leptomeningitis contained B-cell, plasma cell, and macrophage infiltrates (Fig. 6C). Viral antigen was further detected within macrophages in the leptomeninges by immunofluorescence (Fig. 6D).

Protective efficacy of GS-441524 against experimental FIP. Previous research has shown that GS-441524 is a small-molecule analog acting as an alternative substrate and RNA chain terminator of viral RNA-dependent RNA polymerase that can be applied to treat cats with FIP (20, 21). To evaluate the treatment efficacy of the candidate drug, GS-441524, after failure of NA protection, 10 cats (cats T1 to T10) were randomly divided into three groups, with four cats (cats T1 to T4) in the virus-infected treatment group, four cats in the mock-infected control group (cats T7 to T10), and two cats in the virus-infected group (cats T5 to T6). Cats in the virus-infected treatment group and virus-infected group were orally inoculated with 1 ml of rQS-79 virus at $1 \times 10^{5.5}$ TCID₅₀/ml. Cats in the mock-infected control group were orally inoculated with 1 ml of

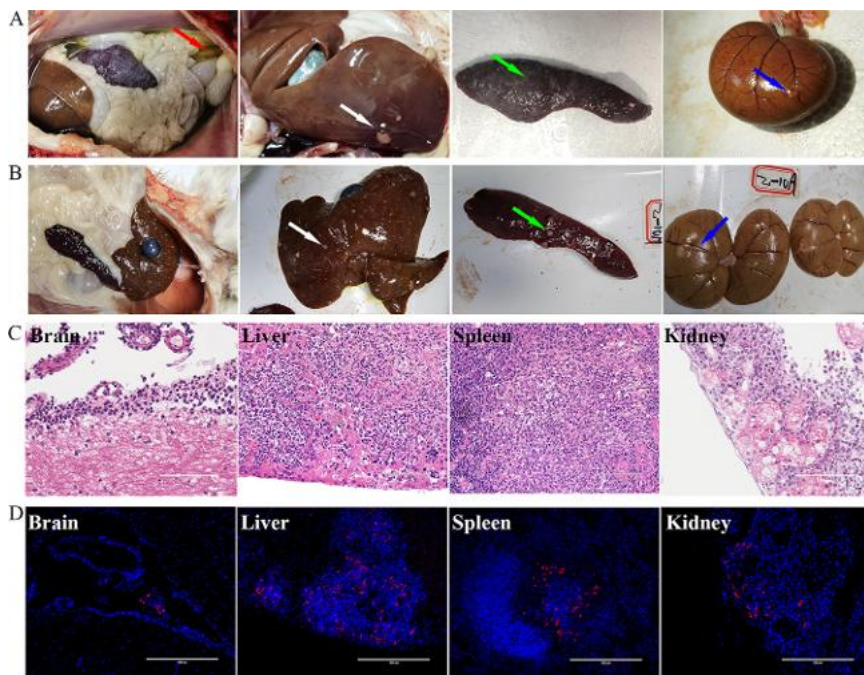


FIG 4 Necropsy and histopathology analysis of rFCoV-infected cats. (A) Gross necropsy of the cat euthanized with effusive FIP. A yellow, protein-rich effusion (red arrow) was observed in the abdomen at necropsy. The liver capsule contained raised and whitish foci, extending into the underlying parenchyma (white arrow). The serosal surface of the spleen was covered with punctate, coalescing fibrinous plaques (green arrow). An obvious granulomatous lesion (blue arrow) was observed on the capsule of the kidney, extending downward into the parenchyma. (B) Gross necropsy of the cat euthanized with non-effusive FIP. The liver capsule contained small punctate raised and whitish foci (white arrow). Several granulomatous lesions (0.2 to 0.4 cm in diameter; green arrow) were observed on the capsule of the spleen, extending downward into the parenchyma. Kidney with granulomatous phlebitis and periphlebitis of a capsular vein (blue arrow). (C) HE stained sections of different tissue sections, including the brain, liver, spleen and kidney. Typical FIP perivascularitis contained B-cell and plasma cell infiltrates in leptomeninges. Distinct focal accumulations of macrophages were observed just under the capsule of the liver, spleen and kidney. These foci were surrounded by dense accumulations of lymphocytes and plasma cells, extending downward into the parenchyma. (D) In the immunofluorescence images, viral antigen was detected in the FIP lesions of the brain, liver, spleen, and kidney and other tissues by anti-rabbit antiserum against FCoV N protein antibody (red).

DMEM (Fig. 7A). Once clinical signs were apparent for 3 days, cats were treated once per day with 5 mg/kg of body weight GS-441524 for 2 weeks (Fig. 7A). After 2 weeks of treatment, we found that all cats recovered to vigor. The body temperature and clinical status gradually returned to normal after 3 days of treatment with GS-441524 compared to the mock group or rQS-79-infected group (Fig. 7B and C). The average relative weight loss in the treatment group (92.5%) was less severe than that in the rQS-79 group (86.4%) compared to that in the mock group (101.03%) (Fig. 7D). Additionally, all cats survived during the treatment compared to the rQS-79-infected group (Fig. 7E). In the course of the follow-up observation, one cat (cat T2) died 2 weeks after the termination of treatment, and the other three cats survived for more than 2 months (data not shown).

DISCUSSION

Many important questions about FIP pathogenesis remain to be answered. In particular, there has been an apparent lack of an effective vaccine for FCoV (18, 22–25). It is now generally accepted that macrophage infection and antibody-dependent enhancement (ADE) of FIPV seem to cause the failure of FIP vaccine development (26–29).

FIPV is thought to represent a mutation of the commonly circulating FECV within individual animals (30). However, it remains unknown where the FECV sequence mutation

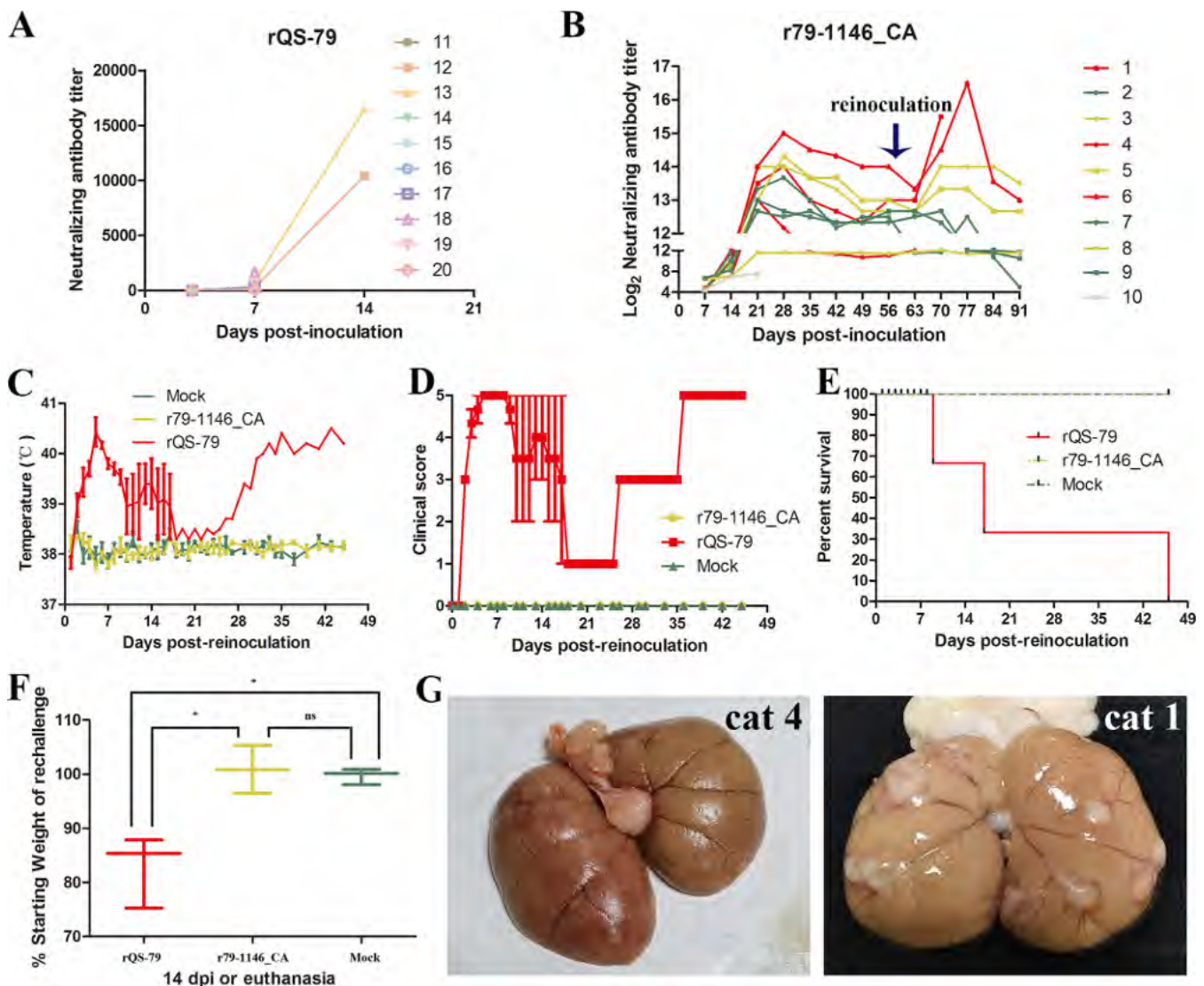


FIG 5 Clinical signs of cats rechallenged with recombinant FCoVs after the first challenge. (A) Induction of FCoV-specific NA titers after oral infection with rQS-79. (B) Induction of FCoV-specific NA titers after oral infection with r79-1146_CA. After 2 months of observation, recombinant FCoVs were reinoculated in r79-1146_CA group cats recovered from the first challenge. (C) The cat body temperatures responded to viral reinoculation over time. (D) The clinical status scores of the cats (in any of the following cases, 1 point will be added) were the following: 1, cats that had a fever ($>39^{\circ}\text{C}$); 2, cats that were anorexic; 3, cats that were lethargic; 4, cats that had weight loss ($>5\%$); and 5, cats that were unable to stand, had severe dyspnea, or were moribund. (E) Percent survival of the cats in the different reinoculation groups. (F) Average relative weight at the time of euthanasia or 14 days after reinoculation. On day 0 before rechallenge, the body weight was 100%. (G) Kidney with granulomatous phlebitis and periphlebitis of a capsular vein in cats 1 and 4. Asterisks indicate significant differences between rQS-79 and mock (*, $P < 0.05$), between rQS-79 and r79-1146_CA (*, $P < 0.05$), and between r79-1146_CA and mock (ns, not significant). The error bars indicate standard deviations.

switches to FIPV (2). To address this core question, virulent FCoVs, attenuated FCoVs, and a corresponding reverse genetic system are urgently needed (1). Furthermore, the challenge of growing FCoV in culture is not limited to pathogenic FIPV but also includes FECV in general. It has been nearly 40 years since the first virulent FIPV 79-1146 was propagated in cell culture (31, 32). During these decades, a series of cell culture-adapted FCoVs have been reported (6, 31, 33–35). However, there is still a lack of highly pathogenic FCoVs that can be propagated in cell culture to produce high titers (3). FCoVs are difficult to grow in cell culture by traditional methods (2, 3). However, the reverse genetics system of constructing full-length cDNAs of field FCoVs enables the production of live virions, which are usually indistinguishable from the parental virions from FIP cats (16).

In the present study, we established a BAC-based reverse genetics system for FCoVs, producing r79-1146_CA and rQS-79, which propagate well in cell culture and

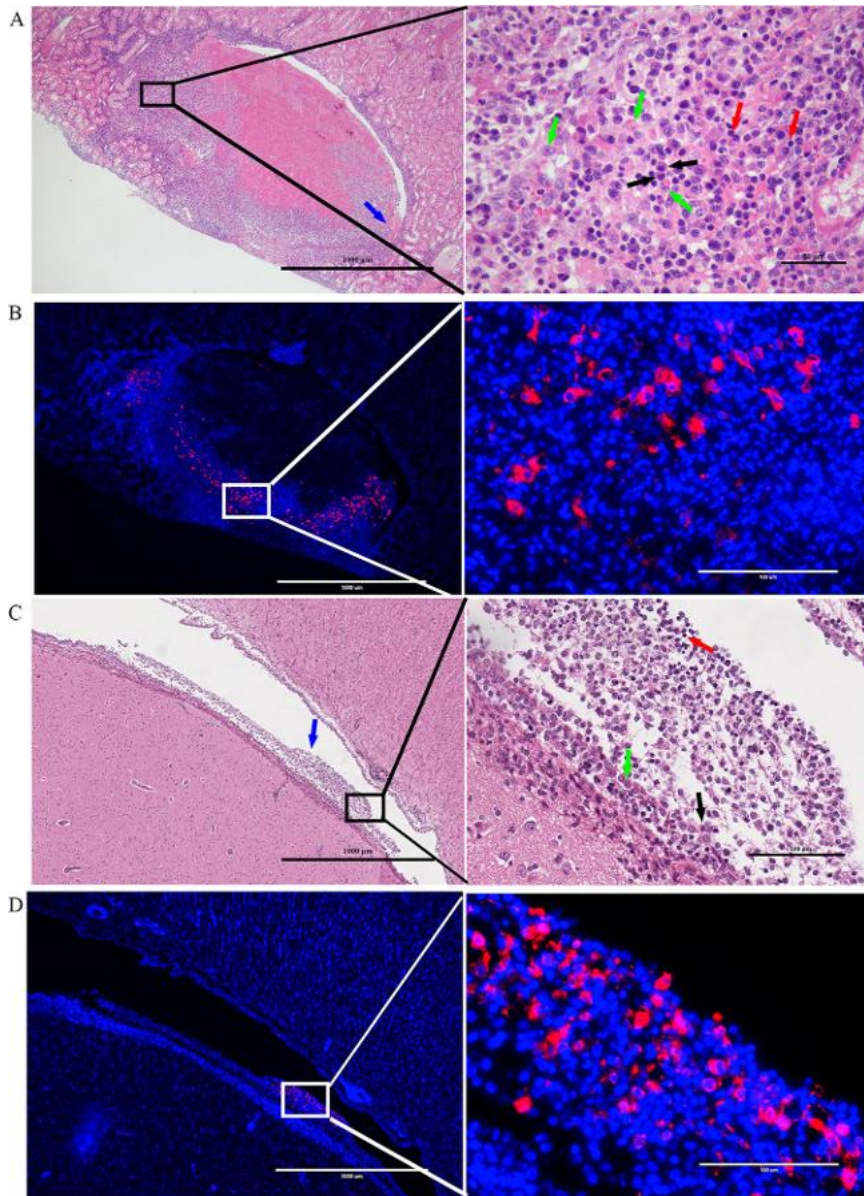


FIG 6 Histopathological examination of kidney and CNS lesions in cats with the rQS-79 reinoculation. (A) A granulomatous phlebitis (blue arrow) and distinct focal accumulations of macrophages (green arrow) were observed. These foci were surrounded by dense accumulations of lymphocytes (red arrow) and plasma cells (black arrow), extending downward into the parenchyma. (B) In the immunofluorescence images, viral antigen was detected in the FIP lesions of kidney by anti-rabbit antiserum against FCoV N protein antibody (red). (C) Typical FIP leptomeningitis (blue arrow) contained B-cell (red arrow), plasma cell (black), and macrophage (green arrow) infiltrates. (D) In the immunofluorescence images, viral antigen was detected in the FIP lesions of kidney by anti-rabbit antiserum against FCoV N protein antibody (red).

exhibit attenuated and virulent biotypes, respectively, when orally inoculated in cats. Although the transcribed products of eukaryocytes (BHK and 293) due to pBCA-QS may be infectious to cats, the application is not convenient (16). There is a lack of stable FCoVs that can cause 100% mortality in cats when applied via the natural route of oral infection rather than intraperitoneal inoculation (3). Here, we first constructed a stable highly virulent FCoV infectious clone, pBAC-QS-79, which will be a great tool to study antivirals and the pathogenesis of FIPV by constructing a series of chimeric mutants between the highly virulent strain and the attenuated strain and related animal experiments. Recombinant rQS-79 can induce FIP (Table 1). Interestingly, Pedersen

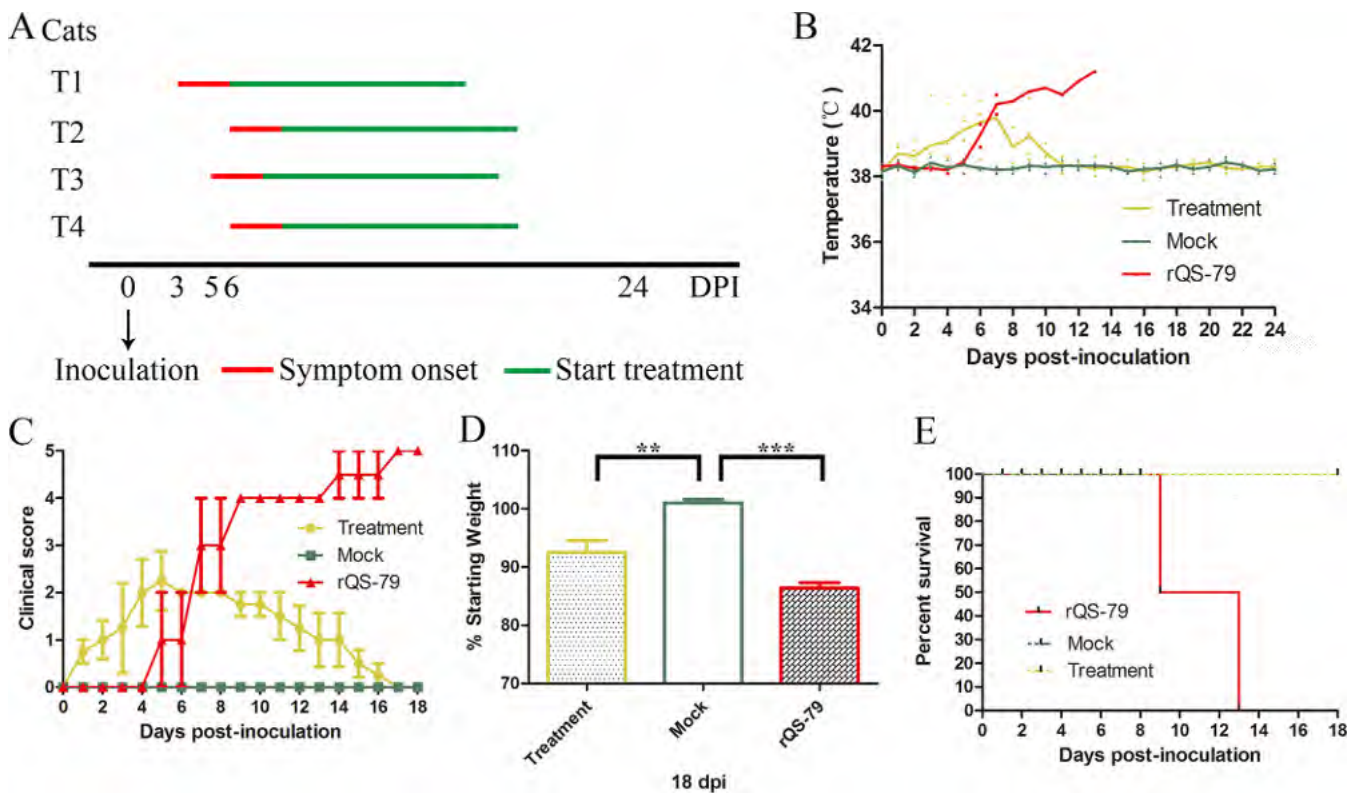


FIG 7 Treatment efficacy of the GS-441524 candidate drug. (A) Schematic diagram showing the treatment strategy for four cats (cats T1 to T4) using the candidate drug, GS-441524. Treatment was started after obvious signs for 3 days. (B) Cat body temperature changed after the treatment. (C) Cat clinical score changed after the treatment. (D) Average relative weight at the time of euthanasia or 18 days after inoculation. On day 0 before challenge, the body weight was 100%. (E) Percent survival of the cats in the animal treatment experiment. Asterisks indicate significant differences between treatment and mock (**, $P < 0.005$), between rQS-79 and mock (***, $P < 0.0005$), and between treatment and rQS-79 (ns, not significant). The error bars indicate standard deviations.

et al. (3, 30) showed that FIPV is present in the feces of experimentally infected cats but that the virus is not infectious when applied via the oral-nasal route. In our study, we found that no live recombinant FCoV existed in feces, but live shed viruses were detected in approximately 10% of infected cats in oral swabs (Fig. 3F and G). These results are consistent with previous reports of difficulty in horizontal transmission of the virus (1, 30).

By rechallenging with recombinant FCoVs in r79-1146_CA group cats recovered from the first challenge, we found that even high NAs did not protect cats from the rQS-79 challenge. This phenomenon may be related to the characteristics of the macrophage-tropic FIPV itself (36). FIPV infection generally causes granuloma formation, which is believed to be caused by a type IV hypersensitivity reaction or the fulminant monocyte activation for the development of FIP vasculitis, followed by a phase of self-sustained granuloma formation (2, 3, 37). Infected monocytes/macrophages can certainly also reach the brain, where they can induce granulomatous vasculitis and perivascularitis (Fig. 6B and C).

Because NAs did not protect cats from the rQS-79 challenge, we further applied the GS-441524 small-molecule analog to treat FIP cats (20, 21). By monitoring the treated cats, we found that GS-441524 relieved cat FIP signs. After approximately 3 days of treatment, the cats regained their appetite. After almost 2 weeks of treatment, all cats fully recovered from FIP clinical signs (Fig. 7B and C). These results indicated that drug therapy is a good option for FIP cats in the absence of an effective vaccine (21).

Despite various approaches attempted, serotype I FIPV QS did not propagate in CRFK or FCWF-4 cell culture systems. Even with reverse genetics, the field QS could not be isolated to grow *in vitro* (Fig. 1B and C). However, the successful construction of the pQS-79 infectious clone demonstrated that the obstacle of the FCoV isolate was

determined by the spike protein (16). The field QS spike protein prevented the recovery of FCoV, which was also attempted in 79-1146_CA. A previous study reported that two point mutations in the spike protein can distinguish FIPV from FECV in 95% of cases (14). In contrast, although both rQS-79 and r79-1146_CA contain the same S gene (S Δ TCD), their pathogenicity was highly virulent and avirulent, respectively, in our animal experiments with rFCoV. These results indicated that the spike protein from serotype II FCoV is not the virulence determinant for FCoV. Terada et al. reported that the type I FCoV strain, C3663, which lacks the 245 amino acids of the N terminus of S, retains its virulence in cats (38). Wang et al. reported that the S gene of PEDV does not determine virulence by constructing a series of chimeric mutants (including the S gene) between the highly virulent strain, BJ2011C, and the avirulent strain, CHM2013 (39). These studies may further support the hypothesis that the S protein is not as important for α -CoV virulence. Whether the S gene from serotype I determines virulence or whether the S gene from serotype II FCoV is part of full virulence will need further study. In conclusion, our study showed that rQS-79 can be rescued in CRFK cells by reverse genetics by introducing a commercially available serotype II FCoV 79-1146 S gene and remains highly virulent, which will provide a great reference for researchers to obtain a highly pathogenic feline coronavirus.

Overall, the establishment of virulent and attenuated full-length cDNA clones and an oral infection model in cats for FCoV, as described here, provides a powerful tool for studying FCoV transmission and pathogenesis as well as evaluating FIP vaccines and therapeutics.

MATERIALS AND METHODS

Cells and viruses. *Felis catus* whole-fetus 4 (FCWF-4) cells were purchased from the ATCC. CRFK cells were kindly provided by Anding Zhang at Huazhong Agricultural University, Wuhan, China. The serotype II FCoV strain 79-1146_CA was initially purchased from the ATCC and passaged over 30 times. The serotype I FCoV strain QS was collected from a euthanized cat with clinically confirmed FIP and donation by the cat owner. All cells and infectious viruses were propagated at 37°C in a 5% CO₂ incubator in Dulbecco's modified Eagle's medium (DMEM; Gibco, Waltham, MA, USA) supplemented with 10% fetal bovine serum (FBS; Gibco, Waltham, MA, USA).

Full-length sequence determination for QS and 79-114679-1146_CA. Ascites were collected from the donated cat with FIP. Total RNA then was extracted from the ascites using TRIzol reagent (Invitrogen, USA), and cDNA was reverse transcribed with reverse transcriptase (TaKaRa, AMV, Japan). The consensus sequences of QS and 79-1146_CA were determined by sequencing (Sangon Biotech, Wuhan, China) of the overlapping PCR products cloned into the vectors in triplicate. The 5' and 3' untranslated region (UTR) sequences were determined by a SMARTer RACE 5'/3' kit (TaKaRa, USA) according to the manufacturer's instructions.

Construction of QS and 79-1146_CA subclones. The FCoV QS genome was divided into eight continuous fragments (A to H), and each fragment was amplified from the total cDNA using specific primers (available upon request). Fragments A, B, C, E, F, G, and H were cloned into pMD18-T (TaKaRa, Japan) using the following endonuclease sites: fragment A was cloned with SfiI and BstXI; fragments B and C were cloned with BstXI; fragment E was cloned with AlwNI; fragment F was cloned with AlwNI and BstXI; fragment G was cloned with BstXI and AlwNI; and fragment H was cloned with AlwNI and SfiI. Moreover, fragment A was cloned to contain a cytomegalovirus (CMV) promoter located at the 5' terminus of the QS genome 5' UTR. Fragment H was cloned to contain the synthesized essential element sequences, including 25A, the HDV RZ sequence (HDV self-cleaving ribozyme sequence), and the BGH transcription terminal signal (Sangon Biotech, Wuhan, China). Fragment D was cloned into the BAC plasmid (preserved in our laboratory) with BstXI and AlwNI to overcome sequence instability (40). To construct the 79-1146_CA subclones, similar methods were used. Specifically, the FCoV 79-1146_CA genome was divided into seven continuous fragments (A to G). Fragments A, B, D, E, F, and G were cloned into pMD18-T. Fragment C was cloned into the BAC to maintain sequence stability. Fragments A and F were cloned with SfiI and BstXI. Fragments B, C, D, and E were cloned with BstXI, and fragment G was cloned with SfiI. All other essential element sequences were constructed using QS subclones. Notably, two natural BstXI sites were removed by introducing the C644T and G6641G mutations, which were maintained as rescue markers.

Assembly of full-length QS and 79-1146_CA cDNA clones. Once all the subclones were determined by sequencing in their corresponding vectors, all subclones were digested with their specific enzymes. Subsequently, all digested products were collected with a gel extraction kit (Omega, USA) and further purified by isopropanol ethanol precipitation. All corresponding fragments of QS or 79-1146_CA were ligated for 2 h at 16°C and transformed into chemically competent DH10B cells (Biomed, China). After determination of all the fragments by bacterial PCR, the positive clones were further determined by restriction fragment length polymorphism with HindIII, and the correct clone was further sequenced (Sangon Biotech, Wuhan, China).

TABLE 2 Details of the oligonucleotide primers used in the experiments

Primer	Sequence
ssDNA-F1	TTAATACGACTCACTATAGGTTGTTGGATTGCTAAGGAGTTTTAGAGCTAGA
ssDNA-F2	TTAATACGACTCACTATAGGATCGAGACTTATGTAAAAGTTTTAGAGCTAGA
ssDNA-R	AAAAGCACCAGCTCGGTGCCACTTTTCAAGTTGATAACGGACTAGCCTTA TTTAACTTGCTATTCTAGCTCTAAAAC
S-QS_F	TATGGTTGTTGGATTGCTAA
S-QS_R	ATGGTGATTGACAAAGTG
S-79_F	CACCTTTGTCAATACACCATGATTGTGCTCGTAACTTG
S-79_R	CACACATACCACGCCATTTTACATAAGTTTCAATTCTATTGAG
79S-F	TCTGGTATGGGTGCTACAGT
79S-R	GTACAGGAGTCTAGGTGTAG

Rescue of the QS and 79-1146_CA full-length cDNA clones. CRFK cells were seeded in a 6-well plate and incubated until 95% confluence, and the recovery of rQS or r79-1146_CA was then performed by transfecting 2 μ g of the corresponding BAC into CRFK cells with 5 μ l of Lipofectamine 2000 (Invitrogen, USA). Once an obvious CPE (~24 h) appeared, the collected virus progenies were determined by IFA and sequenced. The recombinant virus then was further purified from plaques and stored until use at -80°C .

Construction and rescue of recombinant QS with the serotype II FIPV 79-1146_CA S gene. To construct a recombinant QS virus with the serotype II FIPV 79-1146_CA S gene, a CRISPR/Cas9-related method previously reported by our laboratory was used (40). Specifically, two enzyme digestion sites encompassing the QS S gene sequence were selected. Two single guide RNA (sgRNA) templates were produced by the ssDNA-F1, ssDNA-F2, and ssDNA-R target primers (Table 2). Two corresponding sgRNAs transcribed *in vitro* were used to guide the Cas9 nuclease to digest pBAC-QS. The digested products then were purified and recombined with the annealed PCR products constructed by the S-QS_F, S-QS_R, S-79_F, and S-79_R primers (Table 2). The final QS recombinant BAC exchanging the S gene-related sequence with the 79-1146_CA S gene was named pBAC-QS-79. The rescue of pBAC-QS-79 was performed as described above for r79-1146_CA.

Indirect immunofluorescence assays (IFAs). CRFK cells were seeded in 24-well plates. Confluent CRFK cell monolayers were then infected with recombinant FCoV. At 24 h postinfection, cells were fixed in 4% paraformaldehyde for 30 min at room temperature, washed 3 times in phosphate-buffered saline (PBS), permeabilized in PBS containing 0.1% Triton X-100 for 10 min, washed 3 times in PBS, blocked with 2% bovine serum albumin in PBS for 30 min, incubated with rabbit antiserum against FCoV N protein (preparation in our laboratory) for 1 h, washed 3 times, incubated with anti-rabbit IgG secondary antibodies (ABclonal, China) for 45 min, stained with 4',6-diamidino-2-phenylindole (DAPI; Roche, Switzerland) for 5 min at room temperature, washed 3 times in PBS, and examined by fluorescence microscopy.

Viral fluorescent plaque assay. FCWF-4 cells were seeded in 6-well plates. Confluent cells were inoculated with 10-fold serially diluted recombinant virus. After virus adsorption for 30 min, cell monolayers were washed 3 times with DMEM and overlaid with a mixture of 2% low-melting-point agarose and 2 times the concentration of DMEM supplemented with 10% FBS. The overlay was then solidified at 4°C for 5 min. Subsequently, the plates were cultured at 37°C in a 5% CO_2 incubator, and the solidified low-melting-point agarose was removed 2 days postinfection. Cells were then fixed in 4% paraformaldehyde for 30 min, and all the next steps were the same as those described above for IFAs. Finally, the fluorescent plaques were visualized by fluorescence microscopy.

Animal experiments with cats. For the infection of animals with recombinant FCoVs, female cats (from Jiaxiang Huarong Cat Farm, Jining, China) aged 1 to 2 years were used in all experiments. Cats were fed a complete diet (Wapai Oumiao, Jiangsu, China) for adult cats. Each cat was housed in a separate cage at room temperature (25°C). Upon arrival, all cats tested negative for FCoV, feline immunodeficiency virus (FIV), feline panleukopenia virus (FPV), feline herpesvirus (FHV), feline leukemia virus (FeLV), and SARS-CoV-2 by virus-specific RT-PCR on oral and/or rectal swabs using specific primers (available upon request). Approximately 2 months of feeding was performed to acclimatize cats before virus challenge. In the initial experiment, 25 cats were randomly divided into 3 groups, with 10 cats in each virus-infected group and 5 cats in the mock vehicle group. The three groups of cats were kept separate in different catteries in three strictly separated rooms. All virus-infected and mock group cats were orally inoculated with 1 ml of virus (rQS-79 or r79-1146_CA) at $1 \times 10^{5.5}$ TCID₅₀/ml and cell culture DMEM, respectively. During this time, all cats were monitored for body temperature in the inner thigh skin, body weight, clinical state, and percent survival (including euthanization) every day. Viral shedding was measured at 3 and 6 days, and serum NAs were measured every week postinfection. During the animal experiments, dying cats with severe dyspnea and/or inability to stand were euthanized to reduce animal suffering. When the last cat in the rQS-79 group (cats 11 to 20) died from FIP, one cat from the r79-1146_CA group (cats 1 to 10) and one cat from the mock group (cats 21 to 25) were randomly selected for euthanization for a comparative assessment.

After 8 weeks of monitoring from the first animal experiment, nine surviving cats (excluding one for necropsy, cat 10) in the r79-1146_CA group (cats 1 to 9) were roughly divided into three groups according to their average NA titers (each group includes relatively high-, medium-, and low-level NA titers) in

the following experiment. Three cats in the rQS-79 group were reinoculated with 2 ml of rQS-79 virus at $1 \times 10^{5.5}$ TCID₅₀/ml (cats 1, 4, and 6; average NA titers of 8,874), and three cats in the r79-1146_CA group were reinoculated with 2 ml of r79-1146_CA virus at $1 \times 10^{5.5}$ TCID₅₀/ml (cats 3, 5, and 8; average NA titers of 5,761). In addition, three cats in the mock group were reinoculated with 2 ml of DMEM (cats 2, 7, and 9; average NA titers of 7,543). During this experiment, all cats were monitored over 3 months as described above.

The other 10 new naive cats (cats T1 to T10) were randomly divided into three groups, with four cats in the virus-infected treatment group or the mock vehicle group and two cats in the rQS-79 virus-infected group. Cats were inoculated with 1 ml of $10^{5.5}$ TCID₅₀/ml rQS-79 (cats T1 to T6) or DMEM (cats T7 to T10). Four cats (cats T1 to T4) were treated with the GS-441524 candidate drug (synthesized and provided by WuXi Apptec Co., Ltd.) once clinical signs ($>39^{\circ}\text{C}$ or anorexia) were observed for 3 days. In addition, all cats were monitored as described above.

NA titer assays. For the virus neutralization assay, 2-fold dilutions of heat-inactivated (30 min at 56°C) serum were incubated for 1 h at 37°C with $50 \mu\text{l}$ (2×10^3 TCID₅₀/ml) of r79-1146_CA. The mixture of virus and serum then was added to CRFK cells in a 96-well plate. After 2 days of incubation, obvious CPE (syncytia of CRFK cells) was observed, and NA activity was determined by the endpoint dilution method.

Virus detection and live virus titer determination. On days 3 and 6 of recombinant FCoV challenge, oral and anal swabs (Specimen Collection Swab, Shenzhen, China) were taken from all cats and placed into $500 \mu\text{l}$ of culture medium to store at -80°C after vortex blending. To detect shed viruses from the oral and rectal swabs, RT-PCR was conducted using the 79S-F and 79S-R primers (Table 2). Viral titers in the oral and rectal swabs of cats were determined by standard growth curves of viruses. Specifically, CRFK cells were seeded in 96-well plates and grown to approximately 90% confluence. Subsequently, swab culture medium samples ($100 \mu\text{l}$) at 10-fold dilutions were inoculated into CRFK cells in 96-well plates. The viral titers of the recombinant viruses were determined by TCID₅₀ at 48 h post-infection by observing the syncytia of CRFK cells.

Post mortem examination and histopathology. Any cat (severe dyspnea and/or inability to stand) exhibiting extremely sick signs was euthanized for necropsy. When the last cat in the rQS-79 group (cats 11 to 20) died from FIP (at 18 days postinoculation), cats from the r79-1146 and mock groups were randomly selected for euthanasia for necropsy. At necropsy, sections of the lung, liver, brain, cecum, kidney, spleen, and other organs were collected. Cat tissues were fixed in 10% formalin, embedded in paraffin, sectioned, and stained with HE for histopathological examination according to standard procedures for examination by light microscopy.

Rabbit antiserum against FCoV N protein. Antisera were generated against the full-length FCoV 79-1146_CA (GenBank accession no. [MW030109](#)) N protein. For expression of the N protein, the FCoV 79-1146_CA N gene was cloned into the pET-42b prokaryotic expression vector with a C-terminal His tag for protein purification. The initial subcutaneous immunization into Japanese white rabbits was performed with 0.5 mg of N protein suspension mixed with Freund's complete adjuvant 1:1 (vol/vol). Additional boost inoculations were performed three times with 0.5 mg of N protein suspension mixed with Freund's incomplete adjuvant 1:1 (vol/vol). Antiserum was collected with the highest serum titer against the full-length N protein. The titers of antisera evaluated by indirect enzyme-linked immunosorbent assay using the method reported by Han et al. were greater than 12,800 (41).

Immunofluorescence. Paraffin-embedded tissues were sectioned at a thickness of $4 \mu\text{m}$ for immunofluorescence staining. Sections were deparaffinized and rehydrated, and endogenous peroxidases were inactivated with methanol containing 0.3% hydrogen peroxide for 30 min. Antigen retrieval was performed with citrate buffer (pH 6) at 95°C for 30 min. After incubation in blocking solution (5% normal goat serum) for 10 min at room temperature, slides were incubated overnight at 4°C with rabbit antiserum against FCoV N protein. After three washes, sections were incubated with Cy3-labeled goat anti-rabbit IgG at 37°C for 1 h. After three washes, sections were stained with DAPI for 10 min at room temperature. Sections were imaged using fluorescence microscopy. Tissues from cat 21 (mock group cat euthanized for a comparative assessment) and the donated FIP cat (clinically confirmed FIP cat) were used as a negative and positive control, respectively.

Ethics statement. The animal experiments were performed according to protocols approved by The Scientific Ethics Committee of Huazhong Agricultural University (permit number HZAUC-2020-0001). The animal care and maintenance protocols complied with the recommendations detailed in the Regulations for the Administration of Affairs Concerning Experimental Animals made by the Ministry of Sciences and Technology of China.

Informed consent. The related sample collections of donated FIP cat and the following processing were obtained with the cat owner's approval.

Statistical analysis. Statistical analyses were performed using unpaired *t* tests in GraphPad Prism 5. Statistical significance was considered the following: a *P* value of 0.05 was statistically significant, a *P* value of 0.005 was highly significant, and a *P* value of 0.0005 was extremely significant.

Data availability. The full-length genomic sequences determined in this work were deposited in GenBank (accession no. [MW030108](#) and [MW030109](#)).

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We have no potential conflicts of interest related to this study.

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